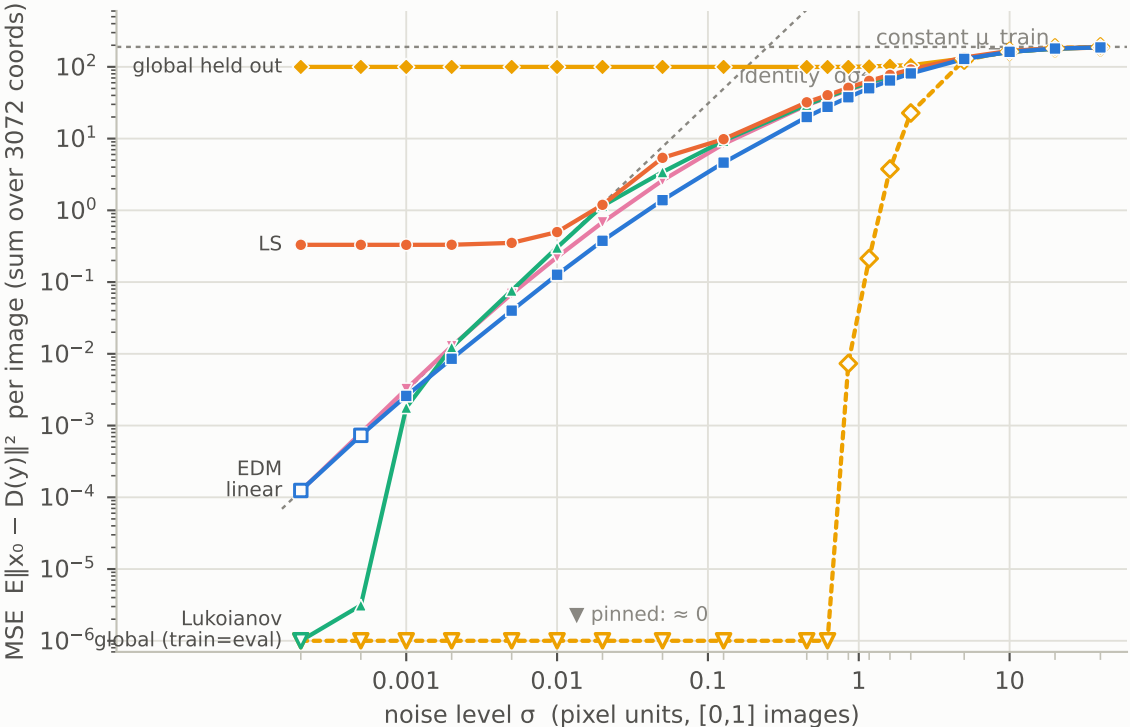


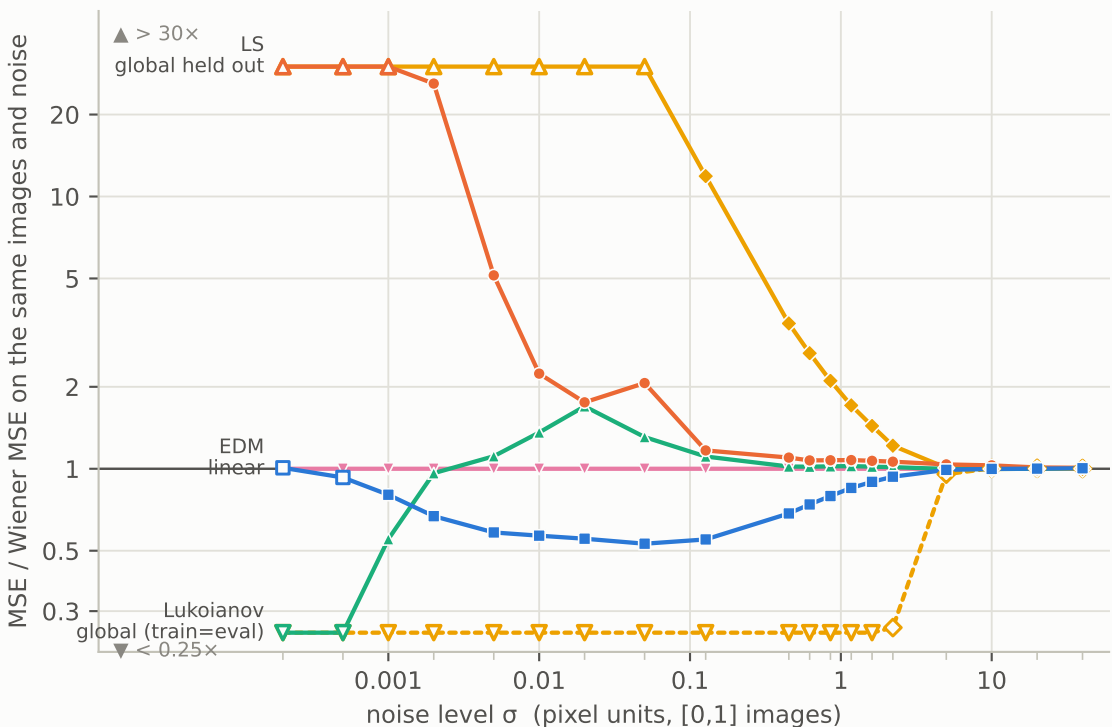
Held out: pool/fit = train[:10000], first 1000 test images, k and τ picked on validation. EDM: trained on 50k; hollow = $\sigma < 0.001$, below its training range. train = eval: pool contains the evaluated image.

MSE vs noise, CIFAR-10



- Linear (Wiener)
- Softmax, global — held out
- Softmax, global — train = eval
- Softmax, local — Lukoianov (Wiener mask)

Relative to linear (< 1 beats Wiener)



- Softmax, local — Kamb & Ganguli LS
- EDM U-net (VP, 50k train)

Lukoianov at $\sigma \leq 0.001 \rightarrow 0$ is an 8-bit artefact: as $\sigma \rightarrow 0$ its Wiener mask shrinks to the diagonal, a per-coordinate nearest value among 10k training pixels, which snaps to the exact 1/255 level once $\sigma \ll 1/255$.
val-selected, by σ ascending: LS k = 1, 1, 1, 1, 1, 1, 1, 1, 3, 7, 7, 9, 11, 15, 19, 31, 31, 31, 15; Lukoianov τ = 0.005, 0.02, 0.1, 0.2, 0.3, 0.5, 0.3, 0.2, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.02, 0.005, 0.5